January 2024, CSE 106
Online on Greedy Algorithms
Time: 40 Minutes

You are given an array of points representing integer coordinates of some points on a 2D plane.

The cost of connecting two points `[xi, yi]` and `[xj, yj]` is the Manhattan distance between them $|x_i - x_j| + |y_i - y_j|$, where $|val|$ denotes the absolute value of `val`.

Return the minimum total cost to connect all the points. All points are connected if there is at least one simple path between any two points.

Input/Output:

The first input line will contain an integer n that indicates the number of given points. The following n lines contain the x and y coordinates of the corresponding points.

The output should be a single line containing the minimum total cost to connect all the points.

Sample Input	Sample Output
5 0 0 2 2 3 10 5 2 7 0	20
3 3 12 -2 5 -4 1	18

Explanation: In the first sample input, we can connect [0, 0] with [2, 2] with cost 4. Similarly, we connect the pairs of points ([2,2], [3, 10]), ([2,2], [5, 2]), and ([5,2], [7, 0]) with costs 9, 3, and 4 respectively. The total cost of connecting all the points is 20.

Submission Guidelines:

Rename your solution as 2205xxx.cpp. Submit the file using the corresponding submission link on Moodle.